



UROP1010 - Final Report

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ABOUT

Research project:

Dynamic Models for Real-Time Autonomous Decision Making in the Real World

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- Autonomous decision-making
- Multi-agent systems
- Stigmergy
- Ad Hoc Teamwork

ABSTRACT

In ad hoc teamwork settings, a focal/ad-hoc agent is required to cooperate with previously unknown teammates while making decisions from limited information. While the introduction of communication grants more information, it also comes with a cost, which the focal agent needs to cautiously consider paying. This project, inspired by how insects employ Stigmergy via pheromones to self-govern, seeks to verify the ability for an ad-hoc agent to interpret the traces left by its teammates in the environment, so as to better communicate at a much cheaper cost. A stigmergic extension to the Level-Based Foraging environment was implemented in the AdLeap-MAS framework to facilitate training convolutional neural network (CNN) policies to interpret traces. In evaluation, the CNN trained with meaningful traces noticeably outperforms the CNN trained with spatially-shuffled traces. An ablation study also points out that the CNN policy benefits from having additional trace values. The results provide initial evidence that the learned policy can interpret spatially meaningful trace information and, therefore, motivate incorporating trace-aware representations into our real-time decision making with dynamic world model and MCTS project.

1 Introduction

Ad hoc teamwork (AHT) is a problem paradigm in which one is tasked with developing ad-hoc agents capable of working with other agents whose identities, purpose and policies are unknown towards a common goal[1]. An ad-hoc agent, furthermore, must be able to adapt its behavior while operating with limited knowledge, which may be caused by partial observation or adversarial teammates. Therefore, the problem is particularly complex due to a severe lack of information.

In mitigating this problem, the ad-hoc agent can directly communicate with teammates to gain information quickly, which, however, often comes with a cost[2]. This causes the ad-hoc agent to consider when and what to communicate, as more communication would mean more computation workload. This project proposes an alternative means of implicit communication derived from a concept called stigmergy. This term refers to a mechanism of coordination used by insects. The main principle is that action performed by an agent leaves a persistent trace in the environment that stimulates the performance of subsequent work[3]. In an environment where all agents perceive traces, a self-organizing structure is generally observed to emerge as tasks are executed in the right order thanks to implicit but natural communication between agents[3].

Inspired by this natural phenomenon, this project ultimately seeks to apply it to ad-hoc teamwork, with the hope to mitigate the drawbacks of direct, traditional communication methods, while achieving on-par information richness. Currently, it aims to verify a premise: *"Can an ad-hoc agent learn to interpret the semantics of stigmergic traces, and, if so, does this improve cooperation with heuristic teammates under partial observability?"*.

The project specifically prioritizes the first part of the question as its result is a prerequisite for a larger system that I intend to develop.

The project extends AdLeap-MAS, an open-source simulator for ad hoc reasoning[4]. The chosen experimental task is Level-Based Foraging (LBF), in which agents with different capability levels cooperate to collect task/food items whose required levels may exceed the capability of individual agents. The implementation introduces a trace-field environment wrapper that employs trace dynamics and converts plain LBF observation to trainable PyTorch tensors that also entail trace values. A CNN model is then trained to be the ad-hoc agent’s policy. The main experimental test compares a model trained with meaningful trace locations against a model trained with the same trace channels spatially shuffled. A second test progressively removes trace channels from the meaningful-trace model to perform an ablation study. These tests aim to verify if the trace semantics is learned and which trace values are most significant in the ad-hoc agent’s interpretation.

The results are positive in signifying the adoption of trace values by the ad-hoc agent. It is a necessary prerequisite to ascertain as the project will propose a potential approach to solve the setbacks of traditional communication in ad-hoc teamwork. Additionally, it continues my interest of applying stigmergic principles to our principal project: "Dynamic Models for Real-Time Autonomous Decision Making in the Real World".

2 Related Work

Ad hoc teamwork

Ad hoc teamwork (AHT) concerns the ability of an agent to cooperate with previously unknown teammates without prior coordination. The central challenge is that the ad hoc agent must infer the behavior and intentions of teammates while operating with incomplete knowledge of their policies. Mirsky et al.[1] survey the field and identify teammate modeling, coordination under uncertainty, and adaptation to unfamiliar teammate policies as major components of AHT. These challenges are directly relevant to the present work, where the ad hoc agent has knowledge of the task distribution but no prior knowledge of the policies of its teammates, which are generated from heuristic behavior templates.

Communication provides one possible mechanism for reducing this uncertainty. Mirsky et al.[1] study communication in AHT and show that an agent can benefit from explicitly reasoning about the timing and content of messages rather than treating communication as universally beneficial. Macke et al.[2] further model communication as a resource with a cost, requiring an agent to balance information obtained through observation against information obtained through communication. These findings suggest that effective coordination is not simply a matter of increasing communication, but also of determining when communication is sufficiently valuable to justify its cost.

The present work investigates an alternative form of coordination based on indirect communication through the environment. Rather than exchanging explicit messages, teammates leave persistent traces that can be observed by the ad hoc agent. This approach is motivated by stigmergy, which provides a mechanism for agents to coordinate through modifications to a shared environment.

Stigmergy and indirect coordination

Stigmergy describes coordination in which an agent’s actions modify a shared environment and the resulting traces influence subsequent actions by other agents[3]. Unlike direct communication, stigmergic coordination does not require agents to exchange explicit messages or maintain direct awareness of one another. This makes stigmergy a potentially useful mechanism for decentralized coordination in environments where agents interact through a shared spatial medium.

Previous work has studied stigmergy as a general mechanism of self-organization, including applications in robotics and other multi-agent settings[3]. However, its use as a learned communication mechanism for ad hoc teamwork remains less explored. In this project, stigmergy is implemented as a structured trace field in the Level-Based Foraging environment. Each trace contains information about its intensity, age, the level of the trace-producing agent, and task-related help and claim signals. This representation allows the ad hoc agent to infer information about previous teammate actions from the spatial and temporal structure of the environment rather than from explicit messages.

CNN policies and learned models

The present study uses a CNN as the ad hoc agent’s policy because the environment is represented as a spatial grid. CNNs are well suited to extracting local spatial patterns that can subsequently be combined into higher-level representations[5]. In this work, the CNN is trained through behavioral cloning on demonstrations generated by a heuristic teammate policy, with an auxiliary value-prediction objective.

The primary question is not whether a CNN can solve Level-Based Foraging in isolation, but whether it can exploit the semantic and spatial structure of stigmergic traces when making decisions. To investigate this,

the study compares a model trained with spatially meaningful traces against a model trained with the same trace values after their spatial locations have been shuffled. If the spatial arrangement of traces contains useful coordination information, the meaningful-trace model should be better able to exploit that information. An additional ablation study examines whether different groups of trace channels contribute differently to the learned policy.

Overall, prior AHT research establishes the challenge of coordination with unfamiliar teammates, while work on communication shows that explicit communication must be used selectively because of its associated cost [1, 2]. Stigmergy provides a complementary mechanism in which coordination information is embedded in the environment itself[3]. This project bridges these areas by investigating whether an ad hoc agent can learn to exploit such environmental traces as an implicit communication channel.

3 Methodology

Environment and observation

The experimental environment is a wrapper around Level-Based Foraging environment in AdLeap-MAS. In a typical training and evaluating instance, an episode uses a 10×10 grid, two agents, and five tasks. The ad hoc agent starts at position (1,1) with level 0.9. Teammates are sampled from L1 and L2 heuristic templates. Parameter knowledge is enabled, type knowledge is disabled, and the vision block is disabled. Agent parameters are sampled in the range [0.5,1.0].

The ad hoc policy does not receive a full history or a direct record of teammate policy type; it acts from the current observation and traces. The wrapper maintains a five-field trace tensor over the grid. After each environment step, trace intensity decays by 0.2, trace age increases toward one, and zero-intensity traces are removed. Active traces can diffuse to neighbouring cells with rate 0.3 with respect to intensity. Agents emit or fuse traces at their positions. If a visible task requires a higher level than an agent possesses, the agent produces a help signal; if the agent can satisfy the task level, it produces a claim signal. When traces overlap, help signal has priority over a competing no-help signal. These rules give the trace channels a specific intended semantics rather than treating them as arbitrary additional features.

The final observation is a tensor with shape $9 \times 10 \times 10$. Three channels encode agent information, channels 3–7 encode intensity, age, level, help signal, and claim signal, and the final channel records visibility. Invisible grid positions are masked by the observation conversion utility. The trace channels are therefore both spatial and temporally suggestive: their locations indicate where a teammate acted, while intensity and age provide information about persistence and recency.

CNN policy and training data

The policy network contains two convolutional layers with 32 and 64 filters, ReLU activations, a flattening layer, and a fully connected representation of size 256. From this representation, a policy head produces logits for five actions and a value head produces one scalar return estimate.

Training data consist of 300 episodes generated with the L1 planning heuristic, which is available in AdLeap-MAS[4]. For each transition, the implementation stores the observation, the heuristic action, and a discounted return. Returns are computed backwards with discount factor $\gamma = 0.99$. The policy is trained using cross-entropy loss on the recorded action and mean-squared error on the return:

$$\mathcal{L} = \mathcal{L}_{\text{policy}} + 0.5\mathcal{L}_{\text{value}}.$$

The Adam optimiser uses learning rate 10^{-3} , batch size 128, and 30 epochs. Thus, the experiment should be understood as behavioural cloning with an auxiliary value-prediction objective. It does not claim that the CNN independently discovered the trace protocol through trial-and-error reinforcement learning.

Meaningful vs. shuffled traces

Among the two separately trained models, the control model preserves the non-trace observation channels and the training targets but applies a fixed random spatial permutation to the five trace channels within each training episode. The meaningful-trace model sees the trace at the location where it was generated; the shuffled-trace model sees trace values detached from their original locations. Both models are then evaluated using the same set of 100 scenarios, which reduces variation caused by comparing different task layouts.

At evaluation time, the CNN chooses the action with the largest policy logit. Each episode is capped at 200 steps. The recorded outcomes are total episode reward, number of steps, and a binary success indicator defined in the evaluation utility as reward greater than zero. The supplied plots report means and, for the ablation plot, 95% normal-approximation confidence-interval margins.

Ablation study

The ablation study evaluates one meaningful-trace model under four input conditions: no trace; basic trace containing intensity and age; level trace containing intensity, age, and agent level; and the full trace containing all five trace fields. The same 100-scenario evaluation procedure is used for all four conditions. This tests whether particular groups of channels are sufficient for the policy’s behaviour, although it is not a retraining-based feature-importance analysis: the model weights remain fixed while input channels are zeroed.

Code availability

The complete implementation, trained model weights, experiment scripts, and instructions for reproducing the experiments are available at: <https://github.com/moitihavoc/adleap-mas-urop1010>.

4 Results

Meaningful and shuffled traces

The meaningful-trace model achieved an average reward of 4.44 per episode, compared with 4.28 for the shuffled-trace model. It also completed episodes in 49.2 steps on average, compared with 64.9 steps for the shuffled model. Since lower step count indicates faster completion, the meaningful model used approximately 15.7 fewer steps per episode, a reduction of about 24.2% relative to the shuffled condition. The reward difference was 0.16 points, or approximately 3.7% relative to the shuffled model.

Table 1: Evaluation on the shared set of 100 scenarios. Values are means read from the supplied evaluation figure.

Condition	Mean reward	Mean steps
Meaningful trace	4.44	49.2
Shuffled trace	4.28	64.9

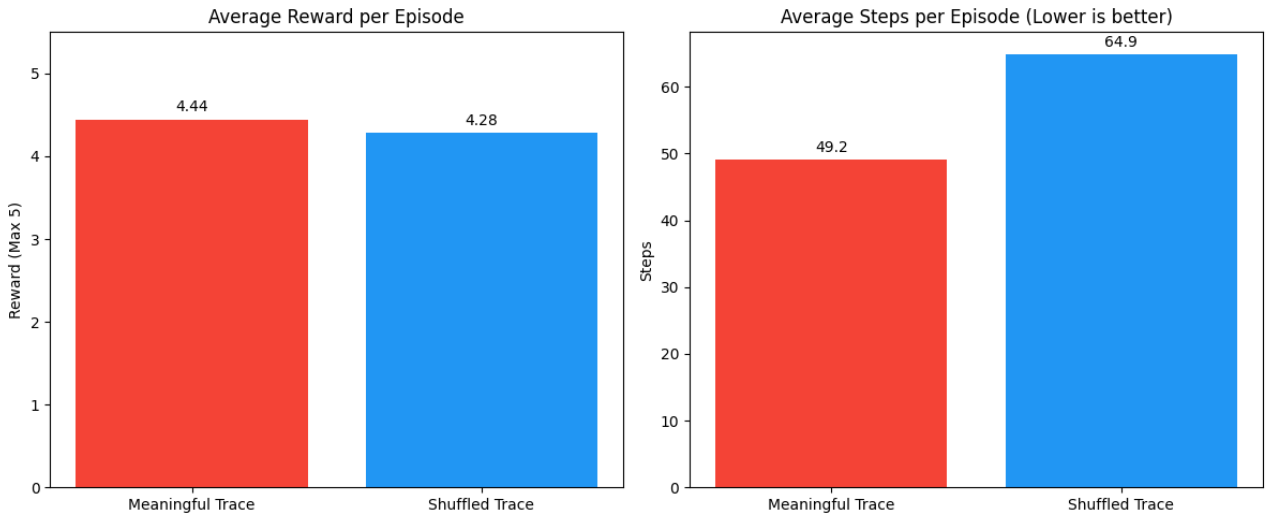


Figure 1: Mean reward and mean episode length for the meaningful-trace and shuffled-trace CNNs. The meaningful-trace model receives the original spatial arrangement of trace values.

Trace ablation

The ablation study produced mean rewards of 4.17 for no trace, 4.18 for basic trace, 4.26 for level trace, and 4.21 for full trace. The corresponding mean step counts were 67.0, 65.2, 61.7, and 65.2. The level-trace condition therefore had the highest observed reward and the shortest observed episodes in this run, while the no-trace condition had the lowest reward and longest episodes. However, the reward differences are small, and the confidence intervals shown in Figure 2 overlap substantially. The result is consequently suggestive rather than conclusive evidence that the level channel is useful.

Table 2: Ablation evaluation on the shared set of 100 scenarios.

Condition	Mean reward	Mean steps
No-trace CNN	4.17	67.0
Basic-trace CNN	4.18	65.2
Level-trace CNN	4.26	61.7
Full-trace CNN	4.21	65.2

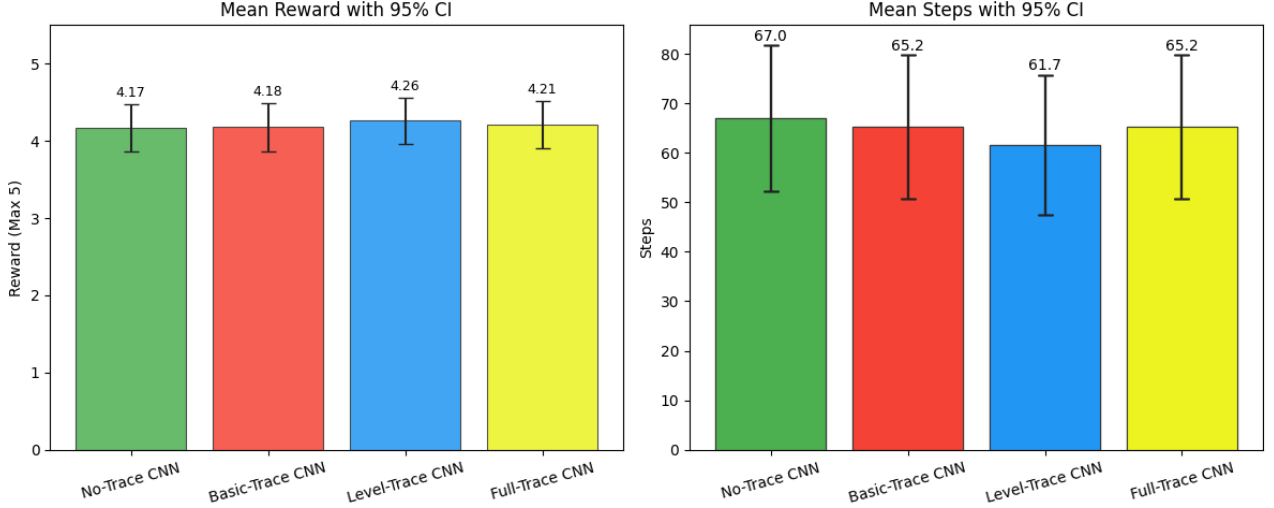


Figure 2: Reward and episode length under progressive removal of trace channels. Error bars represent the 95% confidence-interval margins produced by the evaluation script.

5 Discussion

The main comparison supports the first priority of this project. A model trained on meaningful traces performed better than a model trained on traces whose spatial values had been shuffled, while the non-trace channels and evaluation scenarios were held comparable. The most notable difference was episode length: the meaningful-trace policy reached its outcomes substantially faster. This pattern is consistent with the policy using the relationship between a trace and its grid location, rather than merely exploiting the existence or marginal distribution of trace values.

The evidence is not a proof that the CNN learned human-interpretable semantics in the strongest sense. The meaningful model and shuffled model were trained separately, and the experiment does not include multiple random seeds, a statistical test on paired episode outcomes, a no-trace training baseline, or an evaluation distribution different from the training distribution. Moreover, because the training targets come from an L1 heuristic, the CNN may be learning correlations in the demonstrations rather than a general trace language. The current study also measures aggregate reward and steps, not the specific decisions that demonstrate understanding of help and claim signals.

The ablation results add nuance. Removing all trace channels gave the weakest reward and longest episodes, which is compatible with traces being useful. Nevertheless, adding more channels did not monotonically improve performance: the level-only condition was strongest in the reported means, while the full condition was slightly below it. Possible explanations include redundant or noisy signals, the limited training set, imperfect alignment between the heuristic demonstrations and the evaluation policy, or ordinary sampling variation. Because the model was not retrained for each ablation, these results measure the robustness of an already-trained representation to missing inputs, not the standalone value of each feature.

From an ad hoc teamwork perspective, the implementation demonstrates a practical way to provide indirect coordination under partial observability. Traces carry information forward through the environment without giving the ad hoc agent a direct teammate history. This is useful for the next stage of my research: a learned world model can be trained to predict how traces and task states evolve, and MCTS can then evaluate candidate actions using those predictions. The present results suggest that traces are worth exposing to that model. They do not yet determine the best trace dynamics, the appropriate memory horizon, or whether the learned model will remain useful when teammate policies, grid sizes, or task distributions change.

The next experimental improvements are clear. The study should preserve per-scenario results, report success rates and uncertainty numerically, repeat training and evaluation over several seeds, use paired statistical

comparisons because scenarios are shared, and implement trace diffusion to radiate whole trace instead of only trace intensity as mentioned. A direct interpretability analysis could also intervene on one trace field at a time and inspect the resulting action changes at carefully constructed states. Finally, the CNN should be compared against a no-trace model trained without trace inputs, rather than only receiving zeroed trace channels at test time.

6 Conclusion

This project implemented and evaluated a stigmergic communication mechanism for ad hoc teamwork in Level-Based Foraging. The central comparison found that a CNN trained with meaningful trace locations achieved higher average reward and substantially fewer steps than a CNN trained with spatially shuffled trace values on the same 100 scenarios. This provides initial, implementation-level evidence that an ad hoc policy can learn to use the structure of stigmergic traces under partial observability.

The conclusion is deliberately preliminary. The current experiments verify that the trace representation carries usable information, but they do not yet establish broad generalisation, isolate the causal effect of each semantic field, or demonstrate a full world-model planning system. The most promising next step is to integrate the trace-aware representation into a learned predictive model and evaluate whether MCTS can convert the learned trace interpretation into more robust cooperation with unfamiliar teammates.

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